

Assignment 8: GLMs (Linear Regressions, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

NOTE: For this assignment, you can skip tests for any of the assumptions required to run the models.

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1
#loading packages
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2     3.5.1      v tibble     3.2.1
## v lubridate  1.9.3      v tidyr      1.3.1
## v purrr       1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at /home/guest/EDE_Spring2026
```

```
library(agricolae)
```

```
#working directory  
here()
```

```
## [1] "/home/guest/EDE_Spring2026"
```

```
getwd()
```

```
## [1] "/home/guest/EDE_Spring2026"
```

```
#reading in data  
lakes <- read.csv(  
  here("Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"),  
  stringsAsFactors = TRUE)  
  
#setting date columns  
lakes$sampldate <- mdy(lakes$sampldate)  
  
#2 #building theme  
mytheme <- theme_minimal(base_size = 12) +  
  theme(axis.text = element_text(color = "darkgrey"), legend.position = "bottom",  
        title = element_text(color = "black"))  
theme_set(mytheme)
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question:

Answer: H0: Across all lakes, mean temperatures recorded in July do not change significantly with changing depths
Ha: Across all lakes, mean temperatures recorded in July change significantly with changing depths

4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:

- Only dates in July.
- Only the columns: `lakename`, `year4`, `daynum`, `depth`, `temperature_C`
- Only complete cases (i.e., remove NAs)

5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```

#4
#data wrangling
lakes_July <- lakes %>%
  mutate(month = month(sampledate)) %>% # creating month column
  filter(month == "7") %>% #filter for month 7 / July
  select('lakename', 'year4', 'daynum', 'depth', 'temperature_C') %>% #select columns
  drop_na() #remove NAs

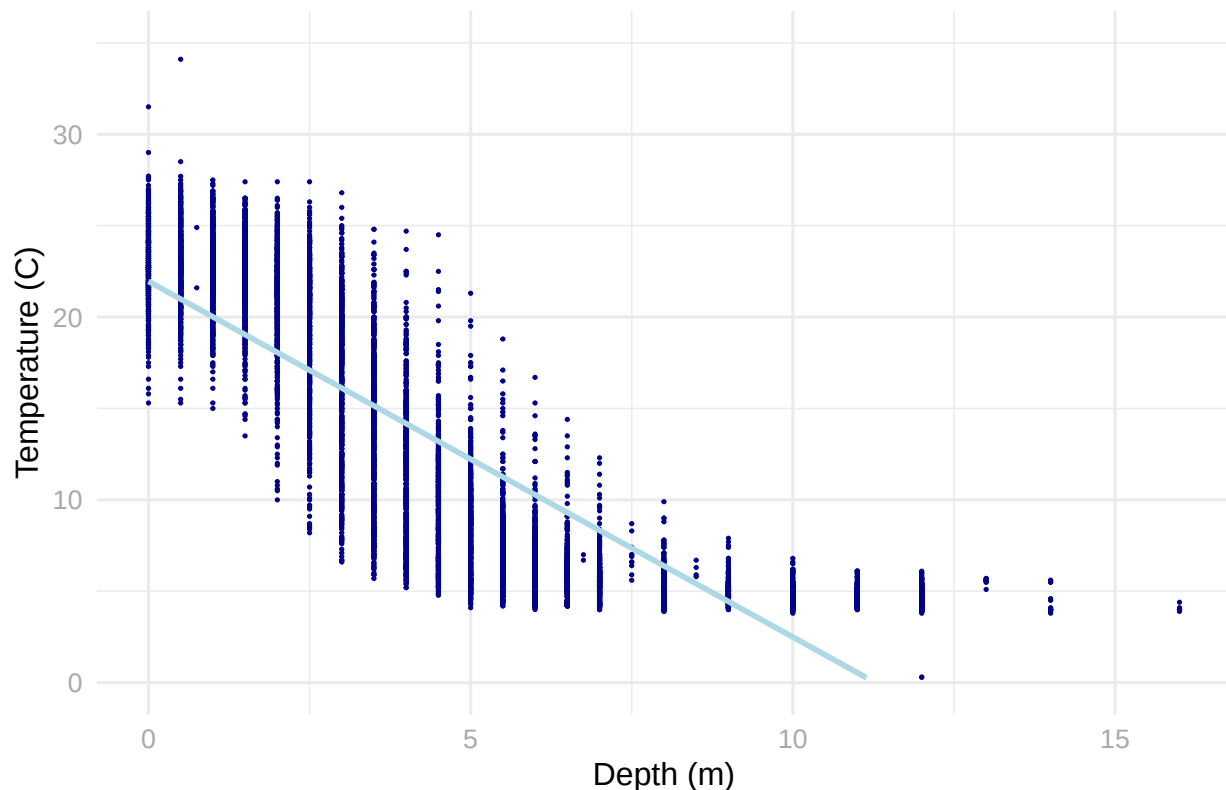
#5 #visualizing relationship
July_plot <- lakes_July %>%
  ggplot(aes(x = depth,
             y = temperature_C)) +
  geom_point(size = 0.25, color = "darkblue") +
  geom_smooth(method = "lm", color = "lightblue") +
  ylim(0,35) +
  labs(x = "Depth (m)",
       y = "Temperature (C)",
       title = "Plotting July Temperatures by Depth Across All Lakes")
print(July_plot)

```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```

Plotting July Temperatures by Depth Across All Lakes



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

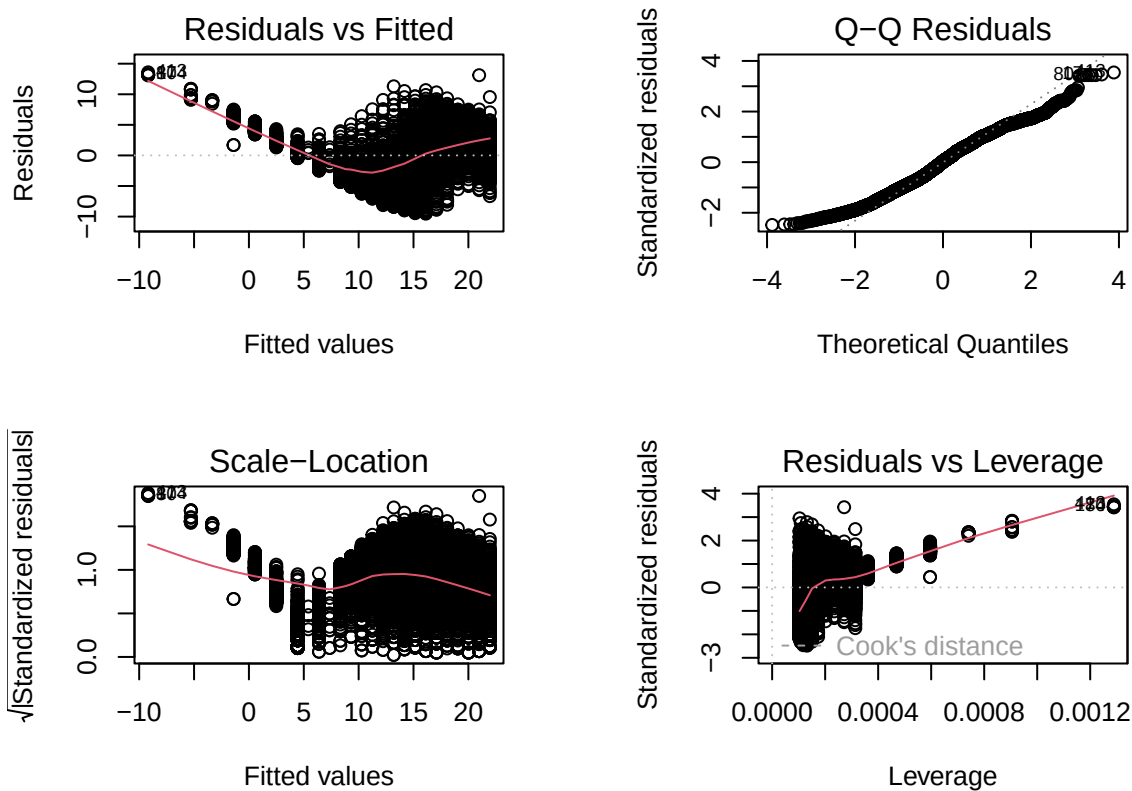
Answer: This suggests that temperatures decrease as depth increases. Although the points seem to suggest some degree of linearity, it is unclear from just viewing the plot whether depth is the main variable that is correlated with temperature. There is a large variation in temperatures at given depths (particularly from 2.5-5m), so it will be helpful to run additional analyses in order to understand the exact relationship between these variables, and whether or not other variables are also relevant.

7. Perform a linear regression to test the relationship and display the results.

```
#7
#running linear regression
lakes_regression <- lm(data = lakes_July, temperature_C ~ depth)
summary(lakes_regression)

##
## Call:
## lm(formula = temperature_C ~ depth, data = lakes_July)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  21.95597    0.06792   323.3  <2e-16 ***
## depth       -1.94621    0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF, p-value: < 2.2e-16

#plotting regression
par(mfrow = c(2,2), mar=c(4,4,4,4))
plot(lakes_regression)
```



```
par(mfrow = c(1,1))
```

- Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: The depth coefficient is -1.94621, indicating a negative relationship between depth and temperature. This means that as depth increases (or, as water gets deeper), temperature decreases. The R-squared value of 0.7387 indicates that changes in depth account for 73.87% of total variability in temperature. This finding is based on 9726 degrees of freedom. The p-value for this coefficient is $2e-16$, meaning that this result fits within the confidence interval to reject the null hypothesis, and is statistically significant. Temperature is predicted to decrease by 1.94621 degrees celcius for every meter increase in depth.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

- Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.

10. Run a multiple regression on the recommended set of variables.

```
#9
#running AIC
lakes_AIC <- lm(data = lakes_July, temperature_C ~ depth + daynum +
               year4)
step(lakes_AIC)

## Start:  AIC=26065.53
## temperature_C ~ depth + daynum + year4
##
##           Df Sum of Sq    RSS   AIC
## <none>                 141687 26066
## - year4    1         101 141788 26070
## - daynum   1         1237 142924 26148
## - depth    1      404475 546161 39189

##
## Call:
## lm(formula = temperature_C ~ depth + daynum + year4, data = lakes_July)
##
## Coefficients:
## (Intercept)      depth      daynum      year4
##   -8.57556   -1.94644    0.03978    0.01134

#10
#running multiple regression
lakes_model <- lm(data = lakes_July, temperature_C ~
                 depth + year4 + daynum)
summary(lakes_model)

##
## Call:
## lm(formula = temperature_C ~ depth + year4 + daynum, data = lakes_July)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994  0.32044
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***
## year4        0.011345   0.004299   2.639  0.00833 **
## daynum       0.039780   0.004317   9.215 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF, p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: Depth, year, and day number are the final set of explanatory variables that the AIC method suggests we use to predict temperature in the multiple regression. No variables were removed because dropping any of the variables would produce a higher AIC value, and thus a worse model. This model has an R-squared value of 0.7412, meaning that it explains 74.12% of observed variance in temperature. This is a marginal improvement from the model that used only depth as an explanatory variable, which accounted for 73.87% of total variability in temperature.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

```
#12 #running ANOVA
#expressed as ANOVA
lakes_anova1 <- aov(data = lakes_July, temperature_C ~ lakename)
summary(lakes_anova1)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      8  21642   2705.2      50 <2e-16 ***
## Residuals    9719 525813     54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#expressed as LM
lakes_anova2 <- lm(data = lakes_July, temperature_C ~ lakename)
summary(lakes_anova2)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = lakes_July)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.769  -6.614  -2.679   7.684  23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    17.6664     0.6501  27.174 < 2e-16 ***
## lakenameCrampton Lake    -2.3145     0.7699  -3.006 0.002653 **
## lakenameEast Long Lake   -7.3987     0.6918 -10.695 < 2e-16 ***
## lakenameHummingbird Lake  -6.8931     0.9429  -7.311 2.87e-13 ***
## lakenamePaul Lake       -3.8522     0.6656  -5.788 7.36e-09 ***
## lakenamePeter Lake      -4.3501     0.6645  -6.547 6.17e-11 ***
```

```
## lakenamTuesday Lake      -6.5972      0.6769  -9.746  < 2e-16 ***
## lakenamWard Lake        -3.2078      0.9429  -3.402  0.000672 ***
## lakenamWest Long Lake   -6.0878      0.6895  -8.829  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

Answer: From the ANOVA model, we observe that the p-value for this analysis is $< 2.2e-16$, indicating that there is a significant difference in mean temperature among the lakes. The linear model provides more in depth statistics including the p-values for each individual lake and relevant significance codes, once again confirming that the difference in mean temperature among the lakes is significant.

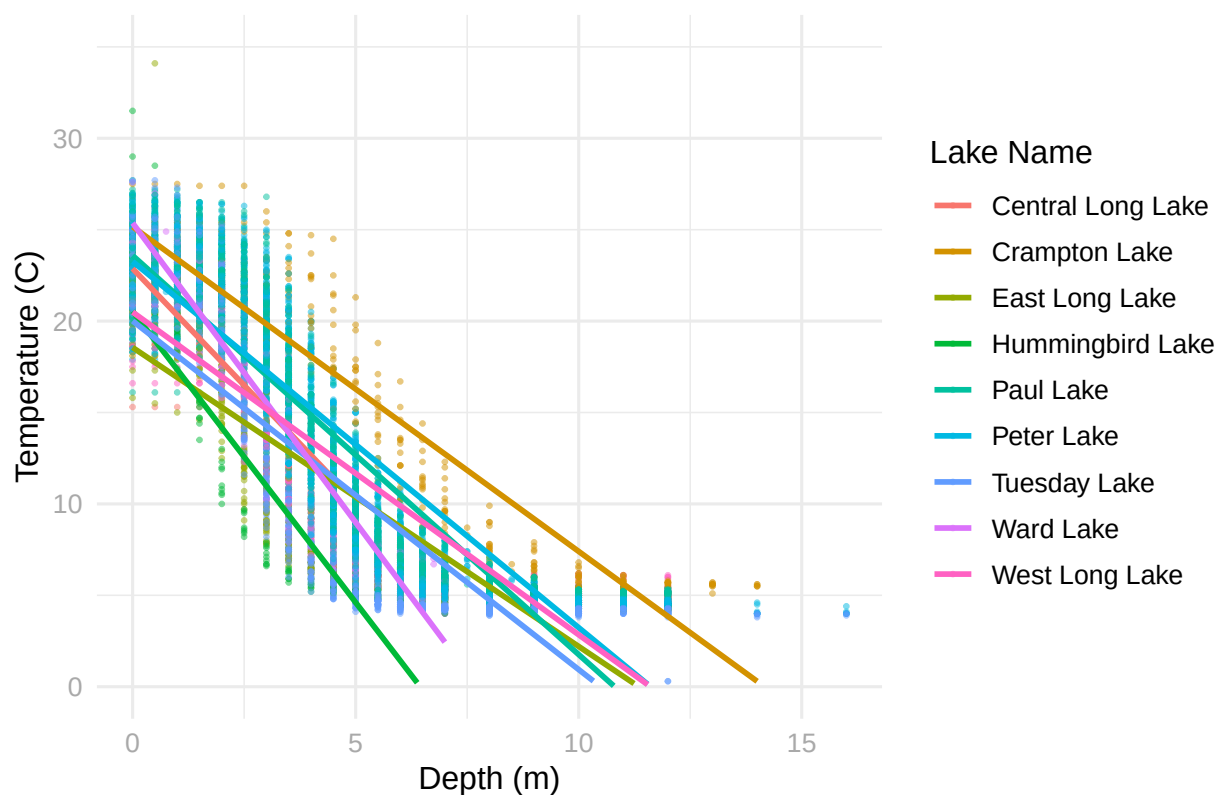
14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
#14.
July_plot <- lakes_July %>%
  ggplot(aes(x = depth,
             y = temperature_C,
             color = lakenam)) +
  theme(mytheme, legend.position = "right") +
  geom_point(alpha = 0.5, size = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  ylim(0,35) +
  labs(x = "Depth (m)",
       y = "Temperature (C)",
       title = "Plotting July Temperatures by Depth Across All Lakes",
       color = "Lake Name")
print(July_plot)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 73 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```

Plotting July Temperatures by Depth Across All Lakes



15. Use the Tukey's HSD test to determine which lakes have different means.

#15

```
TukeyHSD(lakes_anova1)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = lakes_July)
##
## $lakename
##
```

	diff	lwr	upr	p adj
## Crampton Lake-Central Long Lake	-2.3145195	-4.7031913	0.0741524	0.0661566
## East Long Lake-Central Long Lake	-7.3987410	-9.5449411	-5.2525408	0.0000000
## Hummingbird Lake-Central Long Lake	-6.8931304	-9.8184178	-3.9678430	0.0000000
## Paul Lake-Central Long Lake	-3.8521506	-5.9170942	-1.7872070	0.0000003
## Peter Lake-Central Long Lake	-4.3501458	-6.4115874	-2.2887042	0.0000000
## Tuesday Lake-Central Long Lake	-6.5971805	-8.6971605	-4.4972005	0.0000000
## Ward Lake-Central Long Lake	-3.2077856	-6.1330730	-0.2824982	0.0193405
## West Long Lake-Central Long Lake	-6.0877513	-8.2268550	-3.9486475	0.0000000
## East Long Lake-Crampton Lake	-5.0842215	-6.5591700	-3.6092730	0.0000000
## Hummingbird Lake-Crampton Lake	-4.5786109	-7.0538088	-2.1034131	0.0000004
## Paul Lake-Crampton Lake	-1.5376312	-2.8916215	-0.1836408	0.0127491
## Peter Lake-Crampton Lake	-2.0356263	-3.3842699	-0.6869828	0.0000999
## Tuesday Lake-Crampton Lake	-4.2826611	-5.6895065	-2.8758157	0.0000000

## Ward Lake-Crampton Lake	-0.8932661	-3.3684639	1.5819317	0.9714459
## West Long Lake-Crampton Lake	-3.7732318	-5.2378351	-2.3086285	0.0000000
## Hummingbird Lake-East Long Lake	0.5056106	-1.7364925	2.7477137	0.9988050
## Paul Lake-East Long Lake	3.5465903	2.6900206	4.4031601	0.0000000
## Peter Lake-East Long Lake	3.0485952	2.2005025	3.8966879	0.0000000
## Tuesday Lake-East Long Lake	0.8015604	-0.1363286	1.7394495	0.1657485
## Ward Lake-East Long Lake	4.1909554	1.9488523	6.4330585	0.0000002
## West Long Lake-East Long Lake	1.3109897	0.2885003	2.3334791	0.0022805
## Paul Lake-Hummingbird Lake	3.0409798	0.8765299	5.2054296	0.0004495
## Peter Lake-Hummingbird Lake	2.5429846	0.3818755	4.7040937	0.0080666
## Tuesday Lake-Hummingbird Lake	0.2959499	-1.9019508	2.4938505	0.9999752
## Ward Lake-Hummingbird Lake	3.6853448	0.6889874	6.6817022	0.0043297
## West Long Lake-Hummingbird Lake	0.8053791	-1.4299320	3.0406903	0.9717297
## Peter Lake-Paul Lake	-0.4979952	-1.1120620	0.1160717	0.2241586
## Tuesday Lake-Paul Lake	-2.7450299	-3.4781416	-2.0119182	0.0000000
## Ward Lake-Paul Lake	0.6443651	-1.5200848	2.8088149	0.9916978
## West Long Lake-Paul Lake	-2.2356007	-3.0742314	-1.3969699	0.0000000
## Tuesday Lake-Peter Lake	-2.2470347	-2.9702236	-1.5238458	0.0000000
## Ward Lake-Peter Lake	1.1423602	-1.0187489	3.3034693	0.7827037
## West Long Lake-Peter Lake	-1.7376055	-2.5675759	-0.9076350	0.0000000
## Ward Lake-Tuesday Lake	3.3893950	1.1914943	5.5872956	0.0000609
## West Long Lake-Tuesday Lake	0.5094292	-0.4121051	1.4309636	0.7374387
## West Long Lake-Ward Lake	-2.8799657	-5.1152769	-0.6446546	0.0021080

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: From the findings above, Paul Lake and Ward Lake have the same mean temperature (statistically speaking) as Peter Lake, as indicated by the fact that their p-values do not meet the <0.05 confidence interval. None of the lakes have a mean temperature that is statistically distinct from all of the other lakes.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: Another test we could use to determine if Peter and Paul Lake have distinct mean temperatures is a two-sample T-test. This analysis is used to test the null hypothesis that the mean of two samples is equivalent. If the difference in mean temperatures of Peter and Paul Lake is determined to be statistically significant after running the T-test, we can reject the null hypothesis and accept that the two lakes have statistically distinct mean temperatures.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```
crampton_ward_lakes <- lakes_July %>%
  filter(lakename == "Crampton Lake" | lakename == "Ward Lake")

#running t-test
crampton_ward_t_test1 <- t.test(crampton_ward_lakes$temperature_C ~
                                crampton_ward_lakes$lakename)
crampton_ward_t_test1
```

```
##
## Welch Two Sample t-test
##
## data:  crampton_ward_lakes$temperature_C by crampton_ward_lakes$lakename
## t = 1.1181, df = 200.37, p-value = 0.2649
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
##  -0.6821129  2.4686451
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##           15.35189              14.45862
```

Answer: The p-value for this test is 0.2649, meaning that it does not meet the confidence interval to reject the null hypothesis that the difference in mean July temperatures between these two lakes is statistically the same. Thus, there is no statistical difference in mean July temperatures for Ward Lake and Crampton Lake. This is aligned with the Tukey test from questions 15 and 16, whose p-value for Ward Lake-Crampton Lake is 0.9714459, meaning that their mean temperatures are not statistically distinct.